



IUBAT – INTERNATIONAL UNIVERSITY OF BUSINESS AGRICULTURE AND TECHNOLOGY



Founded in 1991 by Md. Alimullah Miyan

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IUBAT School of Computer Science & Engineering (ISCSE)
Department of Computer Science & Engineering

Assignment: Implementation of Compiler Front-End Components

Course Name: Programming Languages and Structures

Course Code: CSC 461

Semester: Summer 2026

Course Instructor: Momtazul Arefin Labib

Lecturer, Dept. of CSE, IUBAT

Assigned: August 4, 2026. Due: September 1, 2026.

Objective:

The objective of this assignment is to implement the fundamental phases of a compiler front-end. Students will progressively develop a program that performs lexical analysis, symbol table construction, parsing, and semantic analysis using attribute grammar.

Your program should read a source code file (or a multiline string) containing a program written in a simplified C-like language.

Task 1: Lexeme Separation

Read the source code and separate it into individual lexemes while preserving their order. Each lexeme should be printed on a separate line.

Sample Input:

Sample Input:

```
int func(int xyz)
{
    xyz=xyz+1;
    return xyz;
}

int main()
{
    int a, b=5;
    for(int i=0;i<=10;i++)
    {
        b=b+func(i);
    }
    return 0;
```

Sample Output:

```
int
func
(
int
xyz
)
{
xyz
=
xyz
+
1
;
return
```



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}

```
xyz  
;  
}  
int  
main  
(  
)  
{  
int  
a  
,  
b  
=  
5  
;  
for  
(  
int  
i  
=  
0  
;  
i  
<=  
10  
;  
i  
++)  
{  
b  
=  
b  
+  
func  
(  
i  
)  
;  
}  
return  
0  
;  
}
```



Task 2: Lexeme Grouping

After separating the lexemes, classify and group them into appropriate lexical categories. At minimum, your program should identify the following categories:

- Keywords
- Functions
- Variables
- Operators
- Others

Sample Input:

Sample Input: <pre>int func(int xyz) { xyz=xyz+1; return xyz; } int main() { int a, b=5; for(int i=0;i<=10;i++) { b=b+func(i); } return 0; }</pre>	Sample Output: Functions: func, main Operators: +, ++, <=, = Variables: a, b, i, xyz Keywords: for, int, return Others: (,), ,, 0, 1, 10, 5, ;, {, }
--	---

Task 3: Identifier Numbering and Lookup Table

Assign a unique identifier number to every variable encountered. The numbering should follow the order of first appearance. Replace variable names with identifier numbers in the grouped output and generate a lookup table.

Sample Input: <pre>int func(int xyz) { xyz=xyz+1; return xyz; } int main() { int a, b=5; for(int i=0;i<=10;i++) {</pre>	Sample Output: Functions: func, main Variables: <1 id> <2 id> <3 id> <4 id> Lookup table: id variable type 1 a int 2 b int 3 I int
---	---



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<pre> b=b+func(i); } return 0; } </pre>	4 xyz int
---	-----------

Task 4: Parse Tree Generation and Semantic Analysis

Implement a parser using the grammar provided below. Your program should:

- Parse the source code.
- Generate a parse tree.
- Perform type checking using attribute grammar.
- Report whether the program passes semantic checking.

Simplified Grammar (BNF):

```

<Program> ::= <Function>

<Function> ::= <Type> id "(" <ParameterList> ")" "{" <StatementList> "}"

<ParameterList> ::= <Parameter>
                  | <Parameter> "," <ParameterList>
                  | ε

<Parameter> ::= <Type> id

<Type> ::= int
        | float
        | char

<StatementList> ::= <Statement> <StatementList>
                  | ε

<Statement> ::= <Declaration>
              | <Assignment>
              | <ReturnStatement>

<Declaration> ::= <Type> id ";"
               | <Type> id "=" <Expression> ";"

<Assignment> ::= id "=" <Expression> ";"

<ReturnStatement> ::= return <Expression> ";"

<Expression> ::= <Expression> "+" <Term>
               | <Expression> "-" <Term>
               | <Term>

```



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```
<Term> ::= <Term> "*" <Factor>
          | <Term> "/" <Factor>
          | <Factor>
```

```
<Factor> ::= id
          | number
          | "(" <Expression> ")"
```

Sample Input	Sample Output
int x; float y; x = 5; y = 2.6 x = y+1	Passed
int x; float y; x = 5; y = 2.6 y = x+1	Type Compatibility Error: Can't assign float to int.
int x; string t; x = 8; t = x;	Type Compatibility Error: Type mismatched.

Submission Details:

- Upload the codes in one or four file, upload the codes in github and share the link.
- Submit A brief report in **HANDWRITTEN** copy to the Course Instructor with a Cover Page.
- You can use both sides of the offset page.
- The assignment should be appropriately stapled. Spiral submission is also allowed.
- The Cover Page should contain all the necessary information.
- The deadline should be strictly maintained.
- **Any Copy and AI Similarity will result in F.**
- You must have to attach the rubrics at the end of your handwritten report



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Assessment Rubrics

Task	Marks	Excellent	Good	Average	Poor	Fail
Task 1: Lexeme Separation	20	18–20: Correctly separates all lexemes in the correct order. Handles keywords, identifiers, literals, operators, and delimiters accurately. Output is complete and properly formatted.	14–17: Minor mistakes (1–2 incorrect or missing lexemes). Output format is mostly correct.	10–13: Several lexemes incorrectly separated or missing, but demonstrates a basic understanding of lexical analysis.	5–9: Many lexemes incorrectly identified or omitted. Output is incomplete or poorly formatted.	0–4: Lexeme separation not implemented or output is largely incorrect.
Task 2: Lexeme Grouping	20	18–20: Correctly classifies all lexemes into Functions, Variables, Keywords, Operators, Constants, and Others. Removes duplicate entries and presents well-organized output.	14–17: Minor classification or duplicate-handling errors. Output is mostly organized.	10–13: Several classification errors, but major groups are correctly identified.	5–9: Many lexemes incorrectly grouped or output lacks organization.	0–4: Grouping not implemented or classifications are mostly incorrect.
Task 3: Identifier Numbering & Lookup Table	25	23–25: Assigns unique IDs consistently based on first occurrence, correctly replaces identifiers with IDs, and generates a complete and accurate lookup table.	18–22: Minor inconsistencies in numbering or lookup table entries. Most replacements are correct.	13–17: Partial implementation with several numbering or lookup errors, but overall approach is evident.	6–12: Identifier numbering is inconsistent and lookup table is incomplete or inaccurate.	0–5: Identifier numbering and lookup table are not implemented or mostly incorrect.



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Task	Marks	Excellent	Good	Average	Poor	Fail
Task 4: Parse Tree & Semantic Analysis	35	32–35: Correctly implements the grammar, generates an accurate parse tree, performs attribute-based type checking, detects semantic errors, and reports clear pass/fail results.	26–31: Minor parsing or semantic checking errors, but overall functionality is correct.	18–25: Parser works for simple inputs, parse tree is partially correct, and type checking is partially implemented.	9–17: Limited parser functionality with incomplete parse tree generation or semantic analysis.	0–8: Parsing and semantic analysis are not implemented or are largely incorrect.



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Justification on PO(b) Attainment:

Assignment Features	Knowledge Profile & Attributes of Complex Engineering Problem	
The solution required an understanding of mathematics and the basic theory of natural science.	K1 & K2	P1- Depth of Knowledge Required
The solution required knowledge about the engineering fundamentals, like the basic structure of a programming language.	K3	
The solution required specialized knowledge about the derivation and implementation of that through computer programs.	K4	
Instead of immediately terminating parsing at the first syntax error (traditional strict compiler behavior), the parser tries to recover from errors, continue parsing, and report multiple errors in one pass.	P2 – Range of Conflicting Requirements	
The solution requires an in-depth analysis of Syntax Analysis. It requires parsing a block of code with multiple condition checks and error detection, instead of an expression (input string). Also, because it uses the last three digits of the ID, the solution for every student will be different.	P3- Depth of Analysis Required	